

**HOW TO USE THIS SET**

20 questions, mixed difficulty, 5 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.**EASY**

How many tunics make up the wall of the GI tract from lumen to exterior?

- A. Three
- B. Four
- C. Five
- D. Six
- E. Two

2.**EASY**

Which structure marks the deep edge of the mucosa and is used to identify it?

- A. Lamina propria alone
- B. Serosa
- C. Meissner's plexus
- D. Muscularis mucosae
- E. Auerbach's plexus

3.**EASY**

Auerbach's (myenteric) plexus is located:

- A. Within the mucosal epithelium
- B. In the serosa
- C. Within the submucosa
- D. Between the inner circular and outer longitudinal layers of the muscularis externa
- E. In the lamina propria

4.**EASY**

Which epithelium lines the esophagus, consistent with its role in protection against abrasion?

- A. Stratified squamous non-keratinized
- B. Pseudostratified ciliated columnar
- C. Simple columnar
- D. Simple cuboidal
- E. Transitional epithelium

5.**MEDIUM****CASE**

A cross-section shows a stellate lumen, thick wall, and skeletal muscle fibres within the muscularis externa.

Which organ and region does this represent?

- A. Sigmoid colon
- B. Stomach fundus
- C. Esophagus, upper third
- D. Esophagus, lower third
- E. Duodenum

6.

MEDIUM

Meissner's (submucosal) plexus primarily controls:

- A. Glandular secretion
- B. Serosal lubrication
- C. Contraction of the muscularis externa for motility
- D. Epithelial turnover
- E. Lymphatic drainage only

7.

MEDIUM

CASE

A field shows an abrupt transition from stratified squamous non-keratinized epithelium to simple columnar epithelium, with mucous glands in the lamina propria just beyond the transition.

What structure is being viewed?

- A. Pylorus
- B. Anorectal junction
- C. Duodenojejunal flexure
- D. Ileocecal valve
- E. Esophagogastric junction

8.

MEDIUM

Which muscularis externa composition is expected in the middle third of the esophagus?

- A. No muscularis externa is present
- B. Purely skeletal muscle
- C. Cardiac-type muscle
- D. A mixture of skeletal and smooth muscle
- E. Purely smooth muscle

9.

MEDIUM

A serosa (rather than an adventitia) covering a GI segment tells you that the segment is:

- A. Suspended within the peritoneal cavity and covered by mesothelium
- B. Part of the esophagus only
- C. Retroperitoneal
- D. Devoid of a muscularis externa
- E. Lined by stratified squamous epithelium

10.

MEDIUM

CASE

A histology slide shows a thin, incomplete layer of smooth muscle producing local folding of the mucosa, distinctly separate from the thick muscularis externa deeper in the wall.

Which layer is being described?

- A. Adventitia
- B. Lamina propria
- C. Serosa
- D. Muscularis mucosae
- E. Submucosa

11.

MEDIUM

Only the abdominal 2–3 cm of the esophagus is covered by true serosa; the rest is covered by:

- A. Muscularis mucosae
- B. Adventitia, since most of the esophagus lies in the thorax outside the peritoneal cavity
- C. Stratified squamous epithelium
- D. No covering layer at all
- E. Simple squamous mesothelium, identical to serosa

12.

MEDIUM

The lamina propria contributes to which of the following functions, according to the structure-function pairing in this lecture?

- A. First-line immune defence (MALT) and nutrient exchange, supporting the epithelium
- B. Housing Auerbach's plexus
- C. Producing bicarbonate-rich secretions
- D. Peristaltic contraction
- E. Forming the serosal mesothelium

13.

MEDIUM

Which scan-path order is recommended for examining an unlabelled GIT cross-section, from lumen outward?

- A. Muscularis externa → mucosa → submucosa → serosa
- B. Random order, since all tunics look identical
- C. Mucosa → submucosa → muscularis externa → serosa
- D. Submucosa → mucosa → serosa → muscularis externa
- E. Serosa → muscularis externa → submucosa → mucosa

14.

MEDIUM

Extrinsic innervation of the GI tract, as distinguished from the two intrinsic plexuses, includes:

- A. Sympathetic and parasympathetic fibres arriving from outside the gut wall, plus blood supply
- B. Meissner's and Auerbach's plexuses only
- C. Glial cells intrinsic to the submucosa
- D. Only the myenteric plexus
- E. Enteric intrinsic neurons exclusively

15.

HARD CASE

A pathology trainee is shown two unlabelled esophageal cross-sections. Section A has an inner circular and outer longitudinal layer composed entirely of smooth muscle. Section B shows the same arrangement but with scattered skeletal muscle fibres intermixed.

Which section most likely comes from a more proximal (upper) level of the esophagus, and why?

- A. Section B, because skeletal muscle is exclusive to the gastroesophageal junction
- B. Section A, because skeletal muscle only appears distally
- C. Section B, because skeletal muscle content decreases progressively from the upper third (all skeletal) through the middle third (mixed) to the lower third (all smooth), so any skeletal fibres place it above the purely smooth lower third
- D. Neither — esophageal muscle type does not vary by level
- E. Section A, because mixed muscle only appears in the upper third

16.

HARD

A segment of bowel is found at surgery to have no mesothelial covering and to be fixed to the posterior abdominal wall by connective tissue. Which layer explains this fixation, and what does its presence (instead of serosa) imply?

- A. Muscularis mucosae; implies the segment is intraperitoneal
- B. Submucosa; implies no functional significance
- C. Lamina propria; implies increased immune activity
- D. Adventitia; implies the segment lies outside the peritoneal cavity (e.g. retroperitoneal or thoracic)
- E. Muscularis externa; implies impaired motility

17.

HARD

If a biopsy shows glands in the submucosa in addition to the standard four tunics, and the epithelium is stratified squamous non-keratinized, which structure does this most likely represent, and what is the source of those glands?

- A. Rectum; anal glands
- B. Stomach body; gastric glands displaced into the submucosa
- C. Jejunum; Peyer's patch-associated glands
- D. Duodenum; Brunner's glands
- E. Esophagus; esophageal (submucosal) glands

18.

HARD

CASE

A researcher damages Auerbach's plexus selectively in an animal model while leaving Meissner's plexus intact.

Which functional deficit would be expected, based on the roles assigned to each plexus in this lecture?

- A. Complete loss of both motility and secretion equally
- B. Selective loss of serosal lubrication
- C. No functional deficit, since the two plexuses are redundant
- D. Impaired coordination of muscularis externa contraction (motility), with glandular secretion relatively preserved
- E. Loss of glandular secretion with preserved motility

19.

HARD

A histology unknown shows a thick wall with a stellate lumen, but the muscularis externa is composed purely of smooth muscle with no skeletal fibres anywhere in the section. Which conclusion is best supported?

- A. This must be the upper esophagus
- B. This is most consistent with the lower esophagus, since skeletal muscle is confined to the upper third and mixed muscle to the middle third
- C. This is the esophagogastric junction specifically
- D. The stellate lumen rules out esophagus
- E. This cannot be esophagus at all, since esophagus always contains some skeletal muscle

20.

HARD

Why is the muscularis mucosae considered part of the mucosa rather than part of the muscularis externa, despite both being smooth muscle layers with 'muscularis' in the name?

- A. Because the two layers are histologically identical and the distinction is arbitrary
- B. Because it lies within the innermost tunic (deep to lamina propria, above the submucosa) and produces local mucosal folding independent of the main peristaltic wave generated by the muscularis externa
- C. Because only the muscularis externa contains smooth muscle
- D. Because the muscularis mucosae is only found in the esophagus
- E. Because the muscularis mucosae is thicker than the muscularis externa

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	Four
2	D	Muscularis mucosae
3	D	Between the inner circular and outer longitudinal layers of the muscularis externa
4	A	Stratified squamous non-keratinized
5	C	Esophagus, upper third
6	A	Glandular secretion
7	E	Esophagogastric junction
8	D	A mixture of skeletal and smooth muscle
9	A	Suspended within the peritoneal cavity and covered by mesothelium
10	D	Muscularis mucosae
11	B	Adventitia, since most of the esophagus lies in the thorax outside the peritoneal cavity
12	A	First-line immune defence (MALT) and nutrient exchange, supporting the epithelium
13	C	Mucosa → submucosa → muscularis externa → serosa
14	A	Sympathetic and parasympathetic fibres arriving from outside the gut wall, plus blood supply
15	C	Section B, because skeletal muscle content decreases progressively from the upper third (all skeletal) through the middle third (mixed) to the lower third (all smooth), so any skeletal fibres place it above the purely smooth lower third
16	D	Adventitia; implies the segment lies outside the peritoneal cavity (e.g. retroperitoneal or thoracic)
17	E	Esophagus; esophageal (submucosal) glands
18	D	Impaired coordination of muscularis externa contraction (motility), with glandular secretion relatively preserved
19	B	This is most consistent with the lower esophagus, since skeletal muscle is confined to the upper third and mixed muscle to the middle third
20	B	Because it lies within the innermost tunic (deep to lamina propria, above the submucosa) and produces local mucosal folding independent of the main peristaltic wave generated by the muscularis externa

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology LG 1, Dr Hero K. Mustafa

Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.